

4.33

$$\begin{aligned}
 & \xrightarrow{x[n]} \boxed{\uparrow 5} \xrightarrow{x_1[n]} \boxed{\downarrow 3} \xrightarrow{x_2[n]} \boxed{\downarrow 5} \xrightarrow{x_3[n]} \boxed{\uparrow 3} \rightarrow y[n] \\
 & \xrightarrow{x[n]} \boxed{\uparrow 5} \xrightarrow{x_1[n]} \boxed{\downarrow 15} \xrightarrow{x_2[n]} \boxed{\uparrow 3} \rightarrow y[n] \\
 & y[n] = \begin{cases} x_3[n/3], & 3|n \\ 0, & 3 \nmid n \end{cases} = \begin{cases} x_1[5n/3], & 3|n \\ 0, & 3 \nmid n \end{cases} = \begin{cases} x_2[n], & 3|n \\ 0, & 3 \nmid n \end{cases} = \begin{cases} x[n], & 3|n \\ 0, & 3 \nmid n \end{cases}
 \end{aligned}$$

4.49

$$\begin{aligned}
 Y_1(e^{j\omega}) &= H_0(e^{j\omega}) X_{10}(e^{j\omega}) = H_0(e^{j\omega}) X_{10}(e^{j2\omega}) \\
 &= H_0(e^{j\omega}) Q_0(e^{j2\omega}) X_{10}(e^{j2\omega}) \\
 &= H_0(e^{j\omega}) Q_0(e^{j2\omega}) \cdot \frac{1}{2} \sum_{k=0}^1 X_{10}(e^{j(\frac{\omega-2\pi k}{2})}) \\
 &= \frac{1}{2} H_0(e^{j\omega}) Q_0(e^{j2\omega}) \sum_{k=0}^1 X(e^{j(\frac{\omega-2\pi k}{2})}) H_0(e^{j(\frac{\omega-2\pi k}{2})})
 \end{aligned}$$

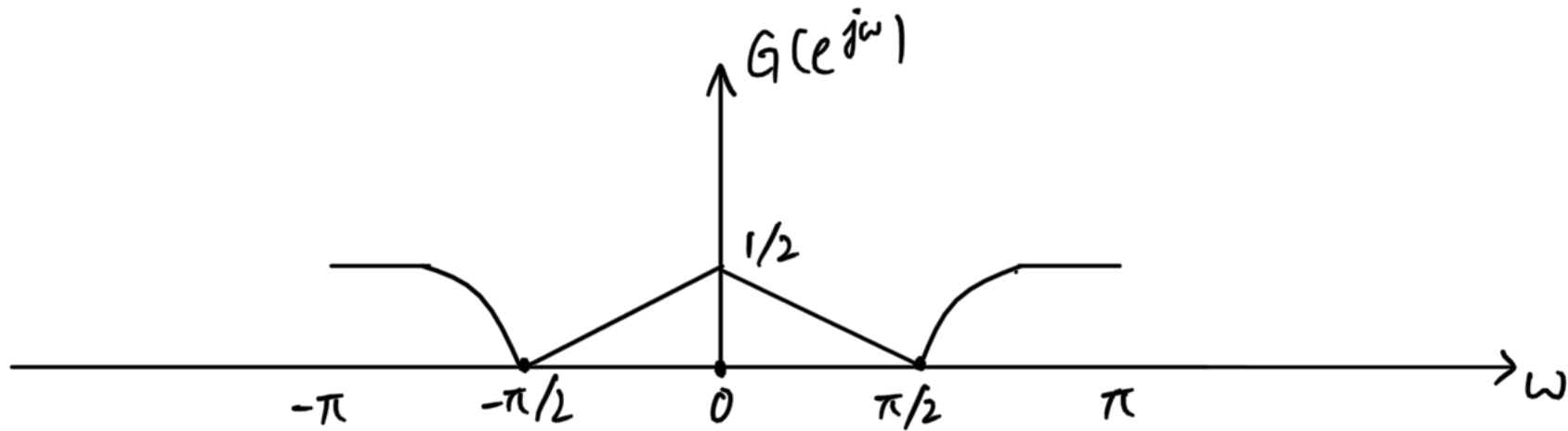
同理可得

$$Y_2(e^{j\omega}) = \frac{1}{2} H_1(e^{j\omega}) Q_1(e^{j2\omega}) \sum_{k=0}^1 X(e^{j(\frac{\omega-2\pi k}{2})}) H_1(e^{j(\frac{\omega-2\pi k}{2})})$$

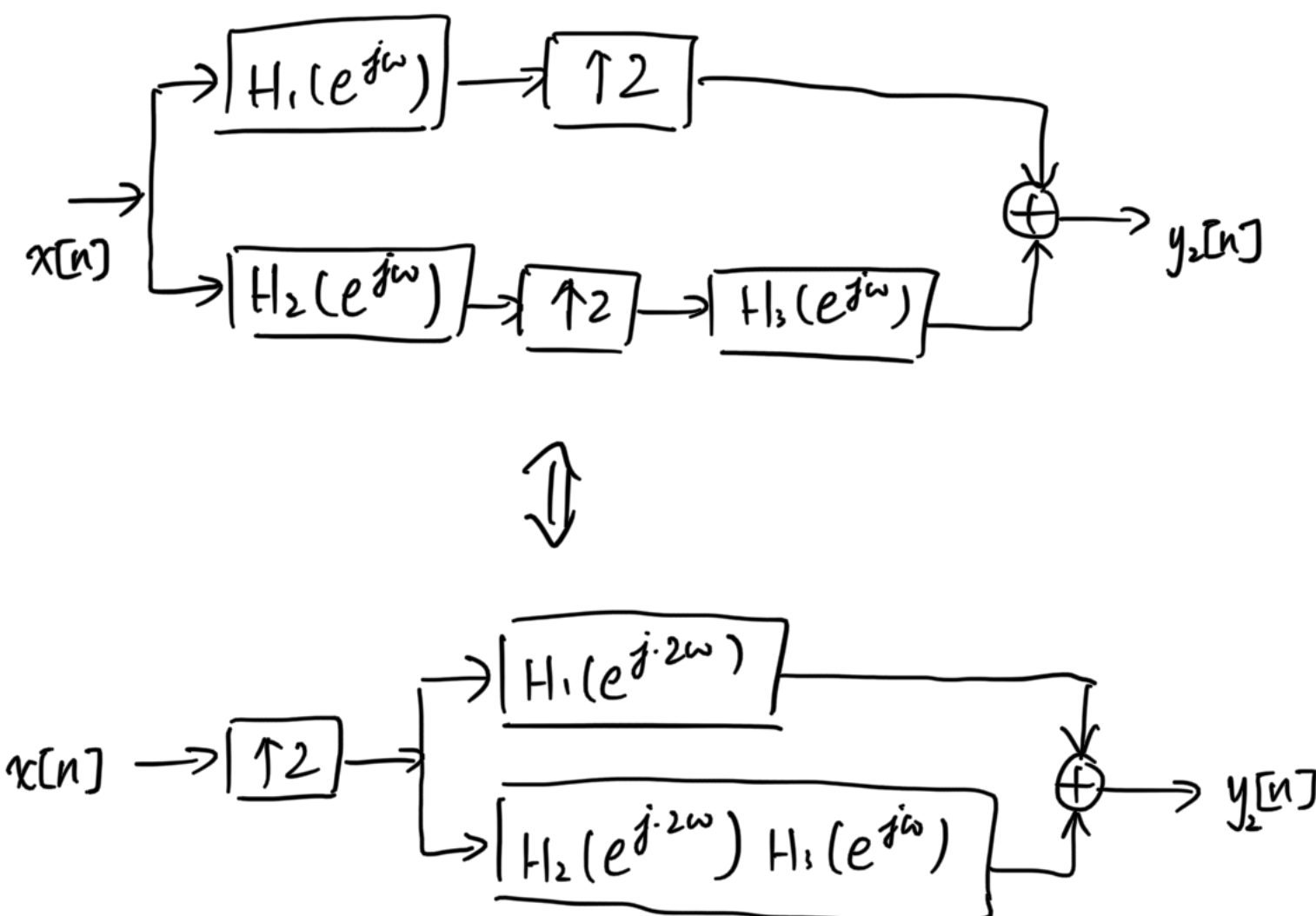
$$\triangleq x[n] = \delta[n] \Leftrightarrow X(e^{j\omega}) = 1$$

$$\text{则 } \sum_{k=0}^1 H_0(e^{j(\frac{\omega-2\pi k}{2})}) = \sum_{k=0}^1 H_1(e^{j(\frac{\omega-2\pi k}{2})}) = 1$$

$$\text{因此 } G(e^{j\omega}) = \frac{1}{2} [H_0(e^{j\omega}) Q_0(e^{j2\omega}) + H_1(e^{j\omega}) Q_1(e^{j2\omega})]$$



4.58 (a)



$$\text{记 } h_i(j) = h_i[j], \text{ 其中 } i=1,2,3, j=0,1,2$$

$$H_1(e^{j2\omega}) \Leftrightarrow h_1^{(1)}[n] = \begin{cases} h(1, n/2), & n=0,2,4 \\ 0, & \text{otherwise} \end{cases}$$

$$H_2(e^{j2\omega}) \Leftrightarrow h_2^{(1)}[n] = \begin{cases} h(2, n/2), & n=0,2,4 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned}
 H_2(e^{j2\omega}) H_3(e^{j\omega}) &\Leftrightarrow h_{23}[n] = \sum_{k=-\infty}^{+\infty} h_2^{(1)}[k] h_3[n-k] \\
 &= \sum_{k=0}^2 h_2^{(1)}[k] h_3[n-2k] \\
 &= \sum_{k=0}^2 h_2[k] h_3[n-2k]
 \end{aligned}$$

$$\triangleq h_1[0:2] = [a, c, e] \Leftrightarrow h_1^{(1)}[0:4] = [a, 0, c, 0, e]$$

$$h_2[0:2] = [b, d, 0] \Leftrightarrow h_2^{(1)}[0:2] = [b, 0, d]$$

$$h_3[0:2] = [0, 1, 0] = \delta[n-1]$$

$$\begin{cases}
 h_{23}[0] = h_2[0] h_3[0] \checkmark \\
 h_{23}[1] = h_2[0] h_3[1] \checkmark \\
 h_{23}[2] = h_2[0] h_3[2] + h_2[1] h_3[0] \checkmark \\
 h_{23}[3] = h_2[1] h_3[1] \\
 h_{23}[4] = h_2[1] h_3[2] + h_2[2] h_3[0] \checkmark \\
 h_{23}[5] = h_2[2] h_3[1] \checkmark \\
 h_{23}[6] = h_2[2] h_3[2] \checkmark \\
 h_{23}[n] = 0, \quad n \neq 0, 1, 2, 3, 4, 5, 6
 \end{cases}$$

$$\begin{cases}
 h_1^{(1)}[0] = h_1[0] \\
 h_1^{(1)}[1] = h_1[1] \\
 h_1^{(1)}[2] = h_1[2] \\
 h_1^{(1)}[n] = 0, \text{ otherwise}
 \end{cases}$$

(b) 图P4.58-1中, 乘法次数 = 5×2 = 10次

图P4.58-2中, 乘法次数 = 3×2+3 = 9次

$$\begin{aligned}
 4.52 \quad H(e^{j\omega}) &= \begin{cases} e^{-j\omega}, & |\omega| < \pi/L \\ 0, & \pi/L < |\omega| \leq \pi \end{cases} \Leftrightarrow h[n] = \frac{1}{2\pi} \int_{-\pi/L}^{\pi/L} e^{-j\omega} e^{j\omega n} d\omega \\
 &= \frac{1}{2\pi} \int_{-\pi/L}^{\pi/L} e^{j\omega(n-1)} d\omega \\
 &= \frac{1}{2\pi} \cdot \frac{1}{j(n-1)} \cdot 2j \sin\left(\frac{\pi(n-1)}{L}\right) \\
 &= \frac{\sin(\pi(n-1)/L)}{\pi(n-1)} = \frac{1}{L} \text{sinc}\left(\frac{n-1}{L}\right)
 \end{aligned}$$

$$\xrightarrow{x_c[n]} \boxed{C/D} \xrightarrow{x[n]} \boxed{\uparrow L} \xrightarrow{x_1[n]} \boxed{h[n]} \xrightarrow{x_1[n]} \boxed{\downarrow L} \rightarrow y[n]$$

$$\begin{aligned}
 y[n] &= x_2[Ln] = \sum_{k=-\infty}^{+\infty} x_1[k] \cdot \frac{1}{L} \text{sinc}\left(\frac{Ln-k-1}{L}\right) \\
 &= \sum_{m=-\infty}^{+\infty} x_1[m] \cdot \frac{1}{L} \text{sinc}\left(\frac{Ln-m-1}{L}\right) \\
 &= \sum_{m=-\infty}^{+\infty} x_1[m] \cdot \frac{1}{L} \text{sinc}\left(n-m-\frac{1}{L}\right) \\
 &= \frac{1}{L} \sum_{m=-\infty}^{+\infty} x_c(mT) \cdot \frac{\sin(\pi(n-m-\frac{1}{L}))}{\pi(n-m-\frac{1}{L})} \\
 &= \frac{(-1)^{n+1}}{L} \cdot \sin\left(\frac{\pi}{L}\right) \sum_{m=-\infty}^{+\infty} x_c(mT) \cdot \frac{(-1)^m}{\pi(n-m-\frac{1}{L})}
 \end{aligned}$$

$$\begin{aligned}
 4.63 \quad (a) \quad E\{e[n]\} &= \int_{-\pi/2}^{\pi/2} e^{-j\omega} \cdot \frac{1}{\Delta} d\omega = 0 \\
 \text{var}\{e[n]\} &= E\{e[n]^2\} - E\{e[n]\}^2 \\
 &= \int_{-\pi/2}^{\pi/2} e^{j\omega} \cdot \frac{1}{\Delta} d\omega = \frac{1}{\Delta a} \cdot 2 \times \left(\frac{\Delta}{2}\right)^2 = \frac{\Delta^2}{12} = \sigma_e^2
 \end{aligned}$$

$$\begin{cases}
 \text{Ree}[n] = E\{e[n] e[n+m]^*\}, \quad n \neq 0 \\
 = \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} e^{j\omega} \left(\frac{1}{\Delta}\right)^* d\omega d\omega' \\
 = E\{e[n] e[n+m]\} = 0 \Rightarrow \text{Ree}[n] = \frac{\Delta^2}{12} \delta[n] \\
 \text{Ree}[0] = E\{e[n]^2\} \\
 = \frac{\Delta^2}{12}
 \end{cases}$$

$$(b) \quad \text{SNR} = \frac{\sigma_x^2}{\sigma_e^2} = \frac{12\sigma_x^2}{\Delta^2}$$

$$(c) \quad E\{x[n] * h[n]\} = E\{x[n]\} * h[n] = 0$$

$$E\{e[n] * h[n]\} = E\{e[n]\} * h[n] = 0$$

$$\begin{aligned}
 E\{(x[n] * h[n])^2\} &= E\left\{\left(\sum_{k=0}^{\frac{\Delta}{2}-1} \frac{1}{2} (a^k - a^{k+\Delta}) x[n-k]\right)^2\right\} \quad E\{(e[n] * h[n])^2\} = \frac{\sigma_e^2}{1-a^2} = \frac{\Delta^2}{12(1-a^2)} \\
 &= E\left\{\left(\sum_{k=0}^{\frac{\Delta}{2}-1} a^{2k} x[n-2k]\right)^2\right\} \\
 &= E\left\{\sum_{k=0}^{\frac{\Delta}{2}-1} \sum_{r=0}^{\frac{\Delta}{2}-1} a^{2(k+r)} x[n-2k] x[n-2r]\right\} \\
 &= \sum_{k=0}^{\frac{\Delta}{2}-1} \sum_{r=0}^{\frac{\Delta}{2}-1} a^{2(k+r)} \sigma_x^2 \delta[2k-2r] \\
 &= \sum_{k=0}^{\frac{\Delta}{2}-1} a^{4k} \sigma_x^2 \\
 &= \frac{\sigma_x^2}{1-a^4}
 \end{aligned}$$

$$\text{SNR} = \frac{\sigma_x^2/(1-a^4)}{\sigma_e^2/(1-a^2)} = \frac{12\sigma_x^2}{\Delta^2}$$

$$(d) \quad f[n] = x[n] e[n]$$

$$E\{f[n]\} = E\{x[n] e[n]\} = 0$$

$$\begin{aligned}
 \text{var}\{f[n]\} &= E\{x[n] e[n]^2\} = E\{x[n]^2\} E\{e[n]^2\} \\
 &= \sigma_x^2 \sigma_e^2 = \frac{\Delta^2}{12} \sigma_x^2 = \sigma_f^2
 \end{aligned}$$

$$\begin{aligned}
 R_{ff}[n] &= E\{f[n] f[n+m]^*\} \\
 &= E\{x[n] x[n+m]^* e[n] e[n+m]^*\} \\
 &= R_{xx}[n] R_{ee}[n] \\
 &= \sigma_x^2 \sigma_e^2 \delta[n] \\
 &= \frac{\Delta^2}{12} \sigma_x^2 \delta[n]
 \end{aligned}$$

$$(e) \quad \text{SNR} = \frac{\sigma_x^2}{\sigma_f^2} = \frac{1}{\sigma_e^2} = \frac{12}{\Delta^2}$$

$$(f) \quad E\{f[n] * h[n]\} = 0$$

$$E\{(f[n] * h[n])^2\} = \frac{\sigma_f^2}{1-a^4} = \frac{\Delta^2}{12} \cdot \frac{\sigma_x^2}{1-a^4}$$

$$E\{(x[n] * h[n])^2\} = \frac{\sigma_x^2}{1-a^4}$$

$$\text{SNR} = \frac{\sigma_x^2/(1-a^4)}{\sigma_f^2/(1-a^4)} = \frac{12}{\Delta^2}$$